



Data Science Career Track

Capstone Three Step 0: Project Ideas

Estimated Time: 3-6 Hours

Overview

For your third capstone, you will follow the [Data Science Method](#). However, the main difference between this capstone project and the second one is that your third capstone project should focus on the topics or techniques identified in the Business Insights Specialization (BIS).

Examples of topics associated with the BIS include, but are not limited to:

- How to allocate marketing budget among different channels to increase product penetration and sales?
- What is the value of our customers now and in X years/months?
- How effective are our marketing strategies?
- How can we characterize our clients?
- What contributes to our clients leaving us?
- What contributes to our clients staying with us?
- What contributes to people becoming our clients?

Examples of projects developed by former mentees include the following:

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Finding Project Candidates and Discussing them with your mentor

1. In collaboration with your mentor, look for business problems and come up with a list of no more than three candidates for your third capstone project, to be later discussed with your mentor. Use the following simple strategy to quickly set aside candidates that do not satisfy any of the following conditions:
 - a. The problem, its relevance, and associated stakeholders can be clearly stated.
 - b. Associated datasets are readily available and obtained from sources that can be clearly identified, and publicly accessible.
 - c. You can visualize approaches--albeit preliminary ones--to model the business problem according to a class of data science problems. For instance, unsupervised, supervised-regression (including time series forecasting), supervised-classification, recommendation systems, etc.
2. The following sources can be used in your search for project candidates: [Google Data Search](#), [Quandl](#), [US Government Open Data](#), [UCI Machine Learning Repository](#), [Kaggle](#), [competitions](#), [Mode Analytics](#). You might also explore [Data is Plural](#), which is an email list that sends newly released and interesting datasets. Do keep in mind, however, that the goal is to look for business problems and not simply looking for datasets. Discuss with your mentor why this is important. Finally, you might also consider using projects from your place of work, either explicitly or implicitly by finding other problems that are similar in nature to the one you are targeting. Discuss with your mentor strategies to accomplish this, and make sure you discuss with your mentor implications of using proprietary data. Based on prior experiences, we recommend that you avoid, if possible, trying to generate data on your own, or 'scraping' data from sources. Be sure you discuss with your mentor what you have in mind if you are considering these options.
3. Once you have settled on no more than 3 candidates, be prepared to pitch each of them to your mentor. Build a simple table with up to three rows, one per candidate, and four columns. The first column is the title of the project, and the remaining three columns address the points in 1 (a-c) above, namely: (a) business problem definition, relevance, and stakeholders; (b) data sources; and (c) anticipated data science approaches to be used.

Student Example

Check out Nick Campanelli's [capstone project](#) to get some inspiration.